Reg. No.														

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2023
First to Third Semester

21BTB103T - BIOLOGY
(For the candidates admitted from the academic year 2022-2023)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40^{th} minute.
(ii) **Part – B** and **Part – C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer **ALL** Questions

Q.	Question	Marks	BL	CO	PO
1.	Eukaryotic chromosomes are composed of (A) DNA + protein (B) DNA + RNA (C) RNA + protein (D) Only DNA	1	2	1	2
2.	Power house of the cells are called (A) Lysosomes (B) Mitochondria (C) Chloroplasts (D) ATP	1	2	1	2
3.	Which phase is the shortest duration in cell cycle? (A) Prophase (B) Metaphase (C) Anaphase (D) S-phase	1	2	1	2
4.	Adult stem cells are found only in (A) Adults (B) Adults, infants and children (C) Children (D) Aged person	1	2	1	2
5.	Cellulose and starch are examples of (A) Monosaccharides (B) Disaccharides (C) Lipids (D) Polysaccharides	1	2	2	2
6.	Lipids are important constituents of (A) Nucleus (B) Ribosomes (C) Mitochondria (D) Biological membranes	1	2	2	2
7.	Sugar molecules in nucleic acid are categorized as (A) Hexase (B) Tetrose (C) Pentose (D) Diose	1	2	2	2
8.	Among these, which one is NOT found in human feces (A) *Pseudomonas sps* (B) *Bactericides sps* (C) *Enterococcus sps* (D) *Bacillus sps*	1	2	2	2
9.	A communicable disease that can be easily transmitted from person to person is called (A) Acute (B) Contagious (C) Allergic (D) Nosocomial	1	2	3	2

10. What infection do antibiotic used to treat? 1 2 3 2
(A) Bacterial infection (B) Viral infection
(C) Fungal infection (D) Parasitic infection

11. Hand virus is responsible for 1 2 3 2
(A) AIDs (B) Bird flu
(C) Dengue (D) Swine flu

12. Which of these biosensors use the principle of heat released during the reaction? 1 2 3 2
(A) Potentiometric biosensors (B) Optical biosensors
(C) Piezo-electric biosensors (D) Calorimetric biosensors

13. For constructing the glucose sensor, which of the following is used as gel? 1 2 3 2
(A) Urea (B) Urease
(C) Acrylamide (D) Polyacrylamide

14. Energy requirement for protein based molecular machine rely on 1 2 4 2
(A) Changes in protein motor units (B) Protein motive force
(C) Breaking of phosphate bond in ATP (D) Breaking of phosphate bond in ADP

15. A biocompatible material is NOT expected to have 1 2 5 2
(A) Cell growth (B) Tissue growth
(C) Adverse host response (D) Bone growth

16. Among these biomaterials, which one is clinically relevant 1 2 5 2
(A) Stainless steel (B) Ti based alloy
(C) Co-Cr-Mo (D) Oxides of iron

17. Compatibility of material with blood is called 1 2 5 2
(A) Histocompatibility (B) Cytocompatibility
(C) Heamocompatibility (D) Tissue compatibility

18. Among these, which material is good for hip joint application? 1 2 5 2
(A) Copper (B) Aluminum
(C) Titanium alloy (D) PLGA

19. Which one is NOT used as a dental biomaterial? 1 2 5 2
(A) Cement (B) Amalgam
(C) PTFE (D) Ti

20. Among these, which one is considered to be new biomaterials for orthopedic implants 1 2 5 2
(A) Polyargletherketones (B) Ti
(C) Hydroxyapatite (D) Amalgam

PART – B (4 × 10 = 40 Marks)
Answer **ANY FOUR** Questions

Q. No.	Question	Marks	BL	CO	PO
21.	Explain with neat diagram of prokaryotic and eukaryotic cell structures.	10	2	1	2
22.	Elaborate the various structures of disaccharides and its various applications.	10	2	2	2
23.	Explain the light dependent reactions in photosynthesis with neat sketch.	10	2	3	2
24.	Cancer is a non-communicable disease. Explain how and write biosensor techniques used to detect the cancer.	10	2	4	2
25.	Discuss the reasons for antibiotic resistance bacteria and its control mechanisms.	10	2	4	2
26.	Elaborate various alloys used in biomaterial applications.	10	2	5	2

PART – C (1 × 15 = 15 Marks)
Answer **ANY ONE** Questions

Q. No.	Question	Marks	BL	CO	PO
27.	Elaborate the various ways to prevent the spread of infectious diseases.	15	2	4	2
28.	Discuss in detail about immune system and its role in the human body.	15	2	3	2

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